

A Novel Model for Detecting Website Defacements Based on BERT and BiLSTM

Xuan Dau Hoang

Faculty of Information Security
Posts and Telecommunications Institute
of Technology
Hanoi, Vietnam
dauhx@ptit.edu.vn

Xuan Nhat Mai

Faculty of Information Security
Posts and Telecommunications Institute
of Technology
Hanoi, Vietnam
nhatmx.b21at147@stu.ptit.edu.vn

Phuong Mai Nguyen

Faculty of Information Security
Posts and Telecommunications Institute
of Technology
Hanoi, Vietnam
mainp.b21at123@stu.ptit.edu.vn

Abstract—Over the past decade, web defacement and related forms of web-based attacks have emerged as significant security threats to enterprise and organizational systems. A single defacement incident can result in severe consequences for the affected party, including immediate disruption of website functionality, reputational damage, and substantial financial losses. To mitigate these risks, various monitoring and detection mechanisms have been developed. However, many of the existing approaches are constrained in scope: some are restricted to static web content, while others can process dynamic content but at the cost of substantial computational overhead. Furthermore, current solutions often suffer from limited detection accuracy and elevated false positive rates, largely due to the inadequate handling of key webpage elements such as plain text content. To address these limitations, this study introduces a novel web defacement detection model leveraging BERT and BiLSTM architectures, with a focus on features extracted from webpage plain text. Extensive experiments conducted on a dataset comprising nearly 70,000 webpages demonstrate that the proposed model achieves superior performance compared to prior approaches, attaining an overall accuracy of 98.61% and an F1-score of 98.70%.

Keywords—Website defacements, Website defacement detection, BiLSTM based defacement detection, BERT based defacement detection.

I. INTRODUCTION

Defacement attacks against web applications and websites constitute a prevalent category of web-based threats, wherein adversaries alter page content to manipulate or distort its visual appearance [1], [2]. Statistical data from Zone-h.org [3] report approximately 500,000 defaced websites worldwide in 2020, with an additional 200,000 cases recorded during the first five months of 2021. An illustrative example is shown in Figure 1, which captures the defacement of the official website www.sohuutritue.gov.vn on April 30, 2021, perpetrated by the hacking group “Overthink1877.”



Fig. 1. Website sohuutritue.gov.vn was defaced on 30th April, 2021

Numerous causes of website and web application defacement have been identified, with the most critical being severe security vulnerabilities in web applications, websites, or their hosting servers that enable adversaries to execute defacement attacks [1], [2]. Among the most prevalent and high-impact weaknesses are Cross-Site Scripting (XSS), SQL Injection (SQLi), remote and local file inclusion, poor account and password management, and the use of outdated software.

Defacement attacks can have serious consequences for website owners. Such incidents may immediately disrupt website functionality, damage organizational reputation, and compromise sensitive data, ultimately leading to substantial financial losses. Due to the widespread and severe implications of defacement incidents, a variety of defensive solutions have been developed. Existing countermeasures can be broadly categorized into three groups:

1. *Vulnerability scanning and remediation tools*, such as Acunetix Web Application Security Testing [4], Trustwave App Scanner [5], and Mister Vulnerability Scanner [6];
2. *Attack monitoring and detection platforms*, including Website Monitoring Service [7], Website Application Monitoring [8], Website Defacement Monitoring [9], and WebOrion Monitor [10];
3. *Defacement detection techniques*, which are the focus of this study and will be further reviewed in Section II.

Nevertheless, many existing countermeasures remain limited in capability: some are applicable only to static web content, while others can handle dynamic content but incur significant computational overhead. In addition, current solutions often exhibit suboptimal detection accuracy and high false positive rates, primarily due to the insufficient processing of critical webpage elements such as plain text content. To overcome these limitations, this paper introduces a novel web defacement detection model that integrates the BERT - a large language model with BiLSTM - a deep learning technique, leveraging plain text features extracted from webpages. The proposed approach is designed to operate effectively on both static and dynamic pages, while improving detection accuracy and reducing false alarms. Specifically, plain text content is extracted from webpages and subsequently processed through BERT and BiLSTM architectures to construct and validate the detection framework using training data.

The primary contributions of this study are summarized as follows:

- Development of a BERT + BiLSTM-based model for web defacement detection using plain text features extracted from webpages;

- Comprehensive experimentation and evaluation of the proposed model on a large-scale dataset containing almost 70,000 webpages.

The remainder of this paper is organized as follows: Section II reviews related studies; Section III details the proposed detection methodology, including preprocessing, training, and detection stages; Section IV presents experimental results and analysis; and Section V concludes the paper with directions for future work.

II. RELATED WORKS

As outlined in Section I, numerous approaches have been proposed for web defacement detection over the past decade. Within the scope of this paper, we focus on representative works belonging to group (3), i.e., methods dedicated to defacement detection employing both simple and advanced techniques. The strengths and weaknesses of these approaches have been analyzed in [11]. Here, we highlight closely related proposals that utilize more sophisticated techniques, including [12]-[16].

In [12] and [13], genetic programming algorithms were applied to construct detection profiles for website defacement. Webpage data were collected using 43 sensors organized into five groups, producing a 1,466-element feature vector. Detection was performed in two phases: during training, data from legitimate webpages were vectorized to generate detection profiles, while during detection, monitored page data were vectorized and compared to these profiles. A significant deviation would trigger an alarm. Although effective, the approach incurs substantial computational costs due to the reliance on genetic programming and high-dimensional feature vectors.

In [11], traditional supervised machine learning techniques were employed for defacement detection. Webpages were vectorized using n-gram and term frequency (TF) methods applied to HTML code, and models were trained on a dataset of 300 defaced and 100 benign pages. Classifiers based on Naïve Bayes and J48 decision trees achieved high accuracy with relatively few false alarms. However, the method was limited by the small dataset size, reducing the reliability of results, and by its narrow focus on HTML code, excluding critical components such as JavaScript, CSS, and embedded images.

To address these issues, [14] proposed a hybrid model combining signature-based detection with machine learning. The signature-based module first scanned HTML code for predefined attack signatures to rapidly detect known defacements, while the machine learning module classified HTML content using trained classifiers. File integrity was further validated with hashing. Experiments on 1,200 defaced and 1,200 normal webpages showed promising results. However, the hashing technique was only effective for static files. Building on this, [15] introduced a multi-layer model integrating machine learning for textual content analysis (HTML, CSS, JavaScript) and hashing for image integrity validation. While effective for detecting textual defacements, the reliance on hashing limited its effectiveness for embedded images, which are critical in many modern webpages.

In contrast, [16] proposed a BiLSTM-based deep learning model that utilized plain text extracted from webpages instead of HTML code. The rationale was that the word distributions in defaced and normal pages differ significantly. BiLSTM was

chosen for its ability to capture long-term dependencies in sequential data, making it highly suitable for text classification. Experiments on a dataset of 96,000 webpages demonstrated strong performance, with 96.04% overall accuracy and a 2.03% false positive rate.

In summary, the main limitations of existing proposals for defacement detection include:

- Excessive computational overhead caused by complex algorithms [12], [13];
- Narrow scope, with many approaches limited to textual or HTML features while inadequately processing CSS, JavaScript, and embedded images, often relying on basic hashing [11]-[15];
- Small experimental datasets, which undermine result generalizability and reliability [11]-[15].

To address these challenges, our approach extends the work in [16] by introducing a BERT-based preprocessing layer prior to the BiLSTM component. Since BERT is pretrained on massive text corpora, it enhances the model's ability to capture semantic relationships among webpage text tokens, thereby improving detection performance. The proposed model is detailed in Section III.

III. PROPOSED WEB DEFACEMENT DETECTION MODEL

A. Overview BERT, DistilBERT and Baseline Models

1) Overview BERT and DistilBERT

BERT (Bidirectional Encoder Representations from Transformers) is a transformer-based natural language processing (NLP) model introduced by Google AI in 2018 [18]. By employing a bidirectional attention mechanism, BERT effectively captures contextual relationships among words within sentences, leading to substantial improvements across a wide range of NLP tasks, including text classification, question answering, and information retrieval. The model is pre-trained on two core objectives, masked language modeling and next sentence prediction, enabling it to better understand linguistic context. Through large-scale pretraining, BERT learns word dependencies in both left-to-right and right-to-left directions. In the proposed framework, BERT is employed for tokenization and embedding of input data. Owing to its training on massive multilingual corpora, BERT is capable of effectively modeling the semantic context of textual content extracted from webpages.

DistilBERT [19], a distilled variant of BERT, was developed to provide a smaller and more computationally efficient model while retaining much of BERT's representational power. Pretrained as a compact, general-purpose language model, DistilBERT can be fine-tuned for downstream tasks with strong performance. Its development was motivated by the need for lightweight models that can support real-time NLP applications, particularly in resource-constrained or edge environments where full-scale BERT models may be impractical due to their high computational cost. In this study, DistilBERT is utilized as a baseline model for comparative evaluation.

2) Baseline Models

For benchmarking purposes, three related detection models were implemented alongside the proposed framework. These models, referred to as baseline models, are constructed using the following techniques:

- word2vec + BiLSTM, as proposed in [16];
- BERT + Fully Connected layers;
- DistilBERT + Fully Connected layers.

It is important to note that in our experimental setup, both BERT and DistilBERT are employed without fine-tuning.

B. The Proposed Detection Model

The proposed web defacement detection model, illustrated in Fig. 2, operates in two stages: (a) a training stage and (b) a detection stage. In the training stage (Fig. 2(a)), a labeled dataset is constructed by extracting textual content from the

HTML files of both defaced and legitimate webpages. The extracted text is then tokenized and embedded using the BERT model, after which the resulting representations are passed to a BiLSTM network [17] for feature extraction and model training. This process yields a classifier capable of distinguishing between defaced and normal webpages.

In the detection stage (Fig. 2(b)), the plain text content of a monitored webpage is first retrieved and subjected to the same preprocessing and feature extraction pipeline as in the training phase. The processed representation is then evaluated by the previously trained classifier to determine whether the monitored page is in a normal state or has been defaced.

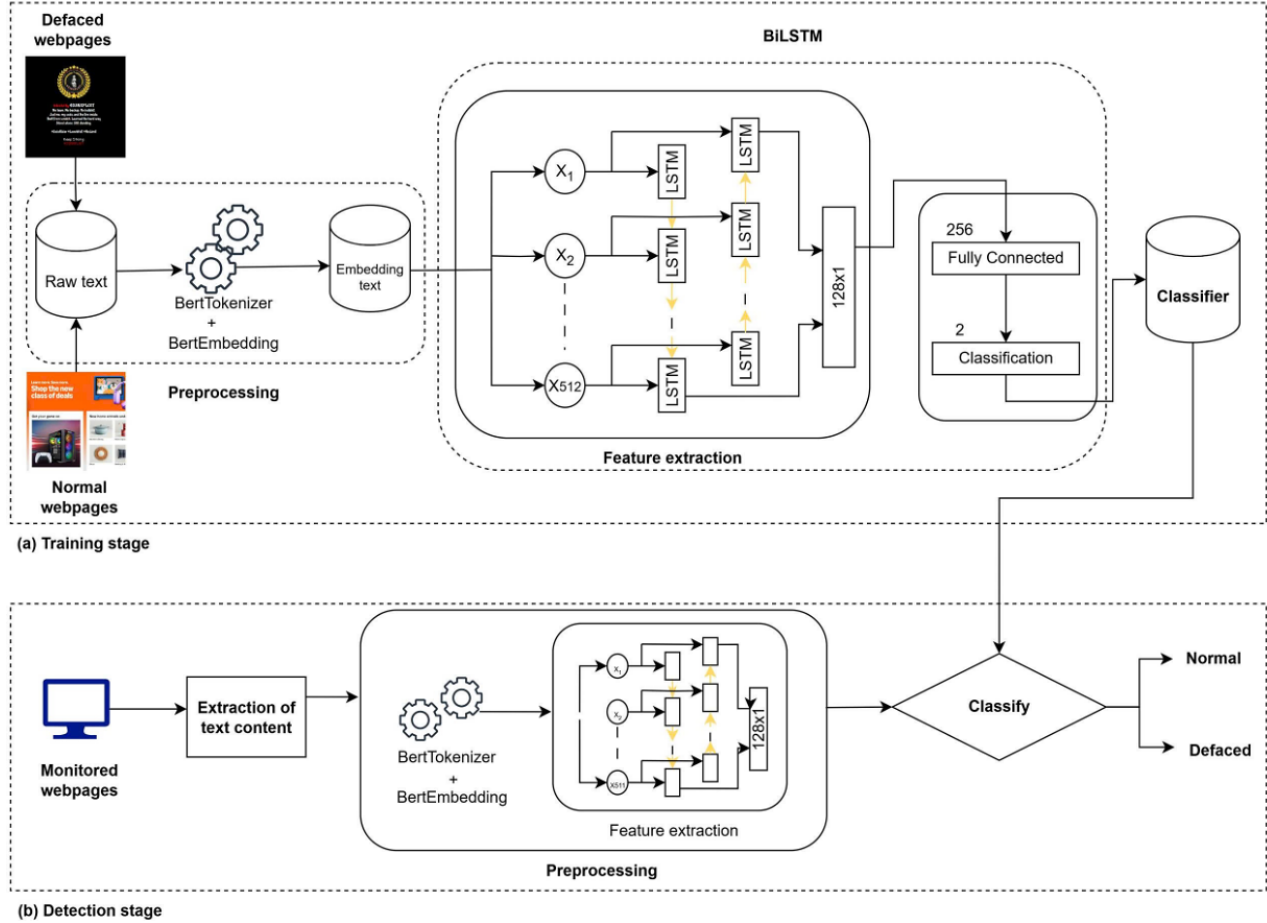


Fig. 2. Architecture of proposed model for web defacement detection: (a) Training stage and (b) Detection stage

C. Data Preprocessing

The preprocessing of the training dataset, comprising both normal and defaced webpages, is conducted in three steps:

- Step 1: Extraction of raw textual content from each webpage, primarily from the title and body sections.
- Step 2: Cleaning and normalization of the extracted text, including conversion of HTML entities (e.g., '&#x2013;', '–') into standard characters and reduction of redundant whitespace to single spaces.
- Step 3: Label assignment, where normal webpages are annotated as 0 (*Normal*) and defaced webpages as 1 (*Defaced*).

D. Model Training

The training process of the proposed BERT + BiLSTM model for webpage text content is carried out as follows:

- Step 1: Text content is tokenized using the BERT tokenizer (*bert-base-uncased*). In this step, special tokens [*CLS*] and [*SEP*] are added, and sequences are normalized to a maximum length of 512 tokens. The tokenizer produces input tensors comprising:
 - + *input_ids*: token indices from the vocabulary;
 - + *attention_mask*: binary indicators distinguishing valid tokens (1) from padding (0);
 - + *token_type_ids*: segment identifiers.
- Step 2: The input tensors are processed by the BERT (*bert-base-uncased*) model. The model outputs the

last_hidden_state, which consists of 768-dimensional embeddings representing the semantic features of each token.

- Step 3: The *last_hidden_state* is passed through a BiLSTM layer with a hidden size of 128 to capture contextual information in both forward and backward directions. The final hidden vector from the forward pass is concatenated with the initial hidden vector from the backward pass, yielding a 256-dimensional representation.

- Step 4: The 256-dimensional vector is passed through a dropout layer ($p = 0.3$) to mitigate overfitting, followed by a fully connected layer that maps the features to two outputs corresponding to the classification labels (0 – *Normal* or 1 – *Defaced*).

The model is trained using the cross-entropy loss function and optimized with Adam (learning rate = $2e-5$, batch size = 32, epochs = 10). Throughout training, performance on the validation set is continuously monitored to prevent overfitting. The model checkpoint with the lowest validation loss is retained as the final classifier.

E. Model Detection

In the detection stage, the monitored webpage is first retrieved and preprocessed following Steps 1 and 2 outlined in Section III.B. The resulting textual content is then processed through the BERT and BiLSTM pipeline as described in Steps 1-3 of Section III.C, producing a 256-dimensional feature vector. Finally, this vector is passed to the trained classifier, which outputs the predicted label: 0 (*Normal*) or 1 (*Defaced*).

F. Performance Metrics

The detection performance of the proposed model is evaluated using six metrics: precision (PRE), recall (REC), false positive rate (FPR), false negative rate (FNR), F1-score (F1), and overall accuracy (ACC). Specifically, PRE quantifies the proportion of correctly identified defacements among all predicted defacements, while REC measures the proportion of correctly identified defacements among all actual defacements. FPR denotes the proportion of normal webpages incorrectly classified as defaced, and FNR represents the proportion of defaced pages incorrectly classified as normal. The F1-score provides the harmonic mean of PRE and REC, reflecting their balance, whereas ACC captures the overall classification accuracy. These metrics are computed using the following formulas:

$$PRE = \frac{TP}{TP + FP} \quad (1)$$

$$REC = \frac{TP}{TP + FN} \quad (2)$$

$$FPR = \frac{FP}{FP + TN} \quad (3)$$

$$FNR = \frac{FN}{FN + TP} \quad (4)$$

$$F1 = \frac{2TP}{2TP + FP + FN} \quad (5)$$

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

where, TP, TN, FP and FN are parameters of the confusion matrix presented in Table I.

TABLE I. THE CONFUSION MATRIX AND ITS TP, TN, FP AND FN PARAMETERS

Predicted Class		Real Class	
		<i>Defaced</i>	<i>Normal</i>
	<i>Defaced</i>	True Positives - TP	False Positives - FP
	<i>Normal</i>	False Negatives - FN	True Negatives - TN

IV. EXPERIMENTS AND RESULTS

A. Experimental Dataset

For the experiments, we employed the dataset of normal and defaced webpages introduced in [16]. The original dataset comprises 57,134 HTML files from normal webpages and 39,100 HTML files from defaced webpages. After removing duplicates and entries with empty textual content, the final dataset consists of 69,912 HTML files, including 32,393 normal pages labeled as 0 and 37,519 defaced pages labeled as 1.

The dataset was randomly partitioned into three subsets:

- Training set (80%): used for model training and parameter optimization;
- Validation set (10%): employed during training to monitor accuracy and adjust parameters, thereby mitigating overfitting;
- Testing set (10%): reserved for evaluating the model's performance after training.

B. Experimental Scenarios and Results

To assess the effectiveness of the proposed detection model in comparison with the baseline models, two evaluation scenarios were designed:

- Test 1: Training and testing the proposed and baseline models using the complete HTML code of webpages;
- Test 2: Training and testing the proposed and baseline models using the extracted pure text content of webpages.

Additionally, we reimplemented the detection model proposed in [16] on the same dataset for comparative analysis.

The performance results are reported in Table II and Table III, corresponding to the use of full HTML code and pure text content, respectively. The findings indicate that models trained on pure text content consistently achieve superior performance compared to those trained on full HTML code. Furthermore, the BiLSTM-based architecture outperforms models based solely on BERT and DistilBERT. Overall, the hybrid BERT + BiLSTM model demonstrates the best performance across both experimental settings.

Table IV presents a performance comparison between the proposed model and prior approaches. The results demonstrate that the proposed BERT + BiLSTM model consistently outperforms earlier methods based on Decision Tree, Random Forest, and standalone BiLSTM algorithms.

C. Discussion

Several key observations can be drawn from the empirical results presented in Tables II and III:

- Detection models utilizing the extracted textual content of webpages consistently outperform those using full HTML code across all evaluated techniques, including BiLSTM, BERT, DistilBERT, and BERT + BiLSTM. This

improvement is attributed to the fact that textual content provides concise and relevant semantic information for attack detection, whereas raw HTML code introduces additional noise.

- All evaluated models, including BiLSTM, BERT, DistilBERT, and BERT + BiLSTM, achieve strong performance with F1-scores exceeding 90%. Among these, the standalone BERT model yields the lowest performance, while the hybrid BERT + BiLSTM model achieves the highest, with F1-scores of 94.31% and 98.70% when trained on textual content.

- The comparative analysis in Table IV further validates the superiority of the proposed BERT + BiLSTM model over prior approaches. Specifically, models based on Decision Tree, Random Forest, and BiLSTM achieve accuracy/F1-scores of 87.33%/84.40%, 93.88%/92.26%, and 97.38%/97.54%, respectively, whereas the BERT + BiLSTM-

based model attains 98.61% for accuracy and 98.70% for the F1-score, representing a significant performance gain.

The superior performance of the proposed BERT + BiLSTM model over previous approaches can be attributed to three key factors. First, the use of plain textual features extracted from webpages is particularly effective for defacement detection, as malicious actors often leave distinctive textual signatures following an attack [16]. Second, employing BERT for data preprocessing is advantageous because its pretraining on large-scale corpora enables it to capture accurate semantic and contextual representations of textual content. Third, the integration of BiLSTM is well-suited for modeling sequential text data, as its bidirectional LSTM units exploit both past and future contextual dependencies, thereby enhancing the model's ability to detect subtle patterns indicative of defacement.

TABLE II. PERFORMANCE METRICS OF PROPOSED AND BASELINE MODELS USING HTML CODE

Detection models	ACC (%)	PRE (%)	REC (%)	F1 (%)	FPR (%)	FNR (%)
Word2vec + BiLSTM	96.63	96.65	96.62	96.63	3.35	3.38
BERT + Fully Connected	90.65	95.34	85.49	90.15	4.18	14.51
DistilBERT + Fully Connected	90.80	94.04	87.14	90.46	5.53	12.86
BERT + BiLSTM + Fully Connected	97.86	99.15	96.55	97.83	0.83	3.45

TABLE III. PERFORMANCE METRICS OF PROPOSED AND BASELINE MODELS USING TEXT CONTENT

Detection models	ACC (%)	PRE (%)	REC (%)	F1 (%)	FPR (%)	FNR (%)
Word2vec + BiLSTM	97.38	98.37	96.72	97.54	1.85	3.28
BERT + Fully Connected	93.94	95.10	93.53	94.31	5.59	6.47
DistilBERT + Fully Connected	94.49	95.37	94.33	94.85	5.32	5.67
BERT + BiLSTM + Fully Connected	98.61	99.33	98.08	98.70	0.77	1.92

TABLE IV. PERFORMANCE COMPARISON OF PROPOSED MODEL AND PREVIOUS PROPOSALS

Detection models	ACC (%)	PRE (%)	REC (%)	F1 (%)	FPR (%)	FNR (%)
Decision tree-based model [11]	87.33	84.4	84.40	84.40	10.67	15.60
Random forest-based model [15]	93.88	93.81	90.76	92.26	4.03	9.24
Word2vec + BiLSTM [16]	97.38	98.37	96.72	97.54	1.85	3.28
BERT + BiLSTM + Fully Connected	98.61	99.33	98.08	98.70	0.77	1.92

V. CONCLUSION

This paper introduces a novel web defacement detection model that integrates BERT and BiLSTM to exploit the semantic and contextual features of textual content extracted from both normal and defaced webpages. The proposed approach achieves high detection accuracy while reducing false alarms, owing to the complementary strengths of BERT in contextual representation learning and BiLSTM in sequential dependency modeling. Extensive experiments on a dataset of nearly 70,000 webpages, including over 32,300 normal and 37,500 defaced samples, demonstrate that the proposed model substantially outperforms existing detection methods across multiple evaluation metrics.

Looking forward, future work will focus on further improving the model by incorporating additional feature sets to (i) enhance detection rates while minimizing false alarms, particularly false negatives, and (ii) reduce computational

overhead during both training and detection, thereby strengthening the practical applicability of the proposed approach for real-world web defacement detection.

ACKNOWLEDGMENT

Authors sincerely thank the Cyber Security Lab, Posts and Telecommunications Institute of Technology, Hanoi, Vietnam for the facility support to complete this research project.

REFERENCES

- [1] Imperva, Website Defacement Attack, <https://www.imperva.com/learn/application-security/website-defacement-attack/>, accessed May 2025.
- [2] Trend Micro, The Motivations and Methods of Web Defacement, https://www.trendmicro.com/en_us/research/18/a/hackivism-web-defacement.html, accessed May 2025.
- [3] Zone-H.org, <http://zone-h.org/archive>, accessed May 2025.
- [4] Acunetix, Acunetix Vulnerability Scanner, <https://www.acunetix.com/vulnerability-scanner/>, accessed in May 2025.

- [5] Trustwave, App Scanner, <https://www.trustwave.com/Products/Application-Security/App-Scanner-Family/App-Scanner-Enterprise/>, accessed in May 2025.
- [6] Misterscanner, Vulnerability Scanner, <https://misterscanner.com>, accessed in May 2025.
- [7] Vietnam Cyberspace Security Technology, VNCS Web Monitoring Service, <https://vncs.vn/en/tin-tuc/detail-vnsc-launches-the-first-online-website-monitoring-service-in-vietnam-281>, accessed May 2025.
- [8] Nagios Enterprises, LLC. Web Application Monitoring. <https://www.nagios.com/solutions/web-application-monitoring/>, accessed May 2025.
- [9] Site24x7. Website Defacement Monitoring. <https://www.site24x7.com/monitor-webpage-defacement.html>, accessed May 2025.
- [10] WebOrion. WebOrion Monitor. <https://www.weborion.io/monitor/>, accessed May 2025.
- [11] X.D. Hoang. A Website Defacement Detection Method Based on Machine Learning Techniques. In *SoICT '18: Ninth International Symposium on Information and Communication Technology*, December 6–7, 2018, Da Nang City, Viet Nam. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3287921.3287975>.
- [12] A. Bartoli, G. Davanzo and E. Medvet. A Framework for Large-Scale Detection of Web Site Defacements. *ACM Transactions on Internet Technology*, Vol.10, No.3, Art.10, 2010. <https://doi.org/10.1145/1852096.1852098>.
- [13] G. Davanzo, E. Medvet and A. Bartoli. Anomaly detection techniques for a web defacement monitoring service. *Journal of Expert Systems with Applications*, 38 (2011) 12521–12530, Elsevier, 2011. <https://doi.org/10.1016/j.eswa.2011.04.038>.
- [14] X.D. Hoang, N.T. Nguyen. Detecting Website Defacements Based on Machine Learning Techniques and Attack Signatures, *Computers* 2019, 8, 35. <https://doi.org/10.3390/computers8020035>.
- [15] X.D. Hoang, N.T. Nguyen. A Multi-layer Model for Website Defacement Detection. In *SoICT'19: Tenth International Symposium on Information and Communication Technology*, December 4 – 6, 2019 | Hanoi - Ha Long Bay, Vietnam. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3368926.3369730>.
- [16] X.D. Hoang, T.H. Nguyen, H.D. Pham. A novel model for detecting web defacement attacks transformer using plain text features. *Indonesian Journal of Electrical Engineering and Computer Science (IJECS)*, vol. 37, no. 1, pp.232-240, 2024. <http://doi.org/10.11591/ijeecs.v37.i1.pp232-240>.
- [17] How to Develop a Bidirectional LSTM For Sequence Classification in Python with Keras, <https://machinelearningmastery.com/develop-bidirectional-lstm-sequence-classification-python-keras/>, accessed May 2025.
- [18] J. Devlin, M.W. Chang, K. Lee, K. Toutanova. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. 2019. arXiv, <https://arxiv.org/pdf/1810.04805>.
- [19] V. Sanh, L. Debut, J. Chaumond, T. Wolf. DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. 2020. arXiv. <https://doi.org/10.48550/arXiv.1910.01108>.
- [20] Alexa, Top 1 million domains. [Available online] <http://s3.amazonaws.com/alexastatic/top-1m.csv.zip>, accessed May 2024.